

Assignment 2: Investment Risk Classification

Machine Learning Pipeline Report

MBAI 5310G 001 | AI Programming | Ontario Tech University | Instructor: Zahra Atf

1. Overview

This report summarises the machine learning pipeline built for Assignment 2. The objective is to predict whether a financial customer is a high investment risk based on personal and financial characteristics. This is a supervised binary classification problem where the model outputs either 0 (No High Risk) or 1 (High Risk). The pipeline covers data loading, cleaning, preprocessing, model training, and evaluation using Logistic Regression.

2. Dataset

The Financial Customer Investment Risk Dataset was used for this assignment. It originally contained 387 rows and 13 columns. After removing 7 duplicate rows the cleaned dataset had 380 rows. Missing values were found in 6 columns and were filled using the median for numerical columns and the most common value for categorical columns.

The dataset is heavily imbalanced. Out of 380 customers only 20 (5.3%) are labelled as High Risk. The remaining 360 customers (94.7%) are labelled as No High Risk. This imbalance directly affects model performance.

Item	Detail
Original rows	387
Rows after cleaning	380 (7 duplicates removed)
Total columns	13
High Risk customers	20 (5.3%)

3. Features and Target Variable

The target variable is High Investment Risk. Eleven feature columns were selected as inputs to the model. The Customer ID column was excluded because it is only an identification number and carries no predictive information.

- Age
- Employment Status

- Annual Income
- Investment Experience Years
- Portfolio Value
- Risk Tolerance
- Number of Investment Products
- Market Volatility Concern
- Financial Literacy Score
- Advisor Contacted
- Previous Investment Loss

4. Methodology

Step 1 Understand the Problem

Identified the business objective, target variable, and type of problem (supervised binary classification).

Step 2 Data Inspection

Loaded the raw dataset and examined shape, data types, missing values, duplicates, and class distribution.

Step 3 Data Cleaning

Removed 7 duplicate rows. Filled missing numerical values with the column median and missing categorical values with the most common value. Saved the cleaned dataset to output/cleaned_dataset.csv.

Step 4 Feature and Target Definition

Defined X with 11 feature columns and y as the target column. Converted the target from text (No/Yes) to integers (0/1).

Step 5 Train and Test Split

Split data 80% for training (304 rows) and 20% for testing (76 rows) using stratified splitting to maintain class proportions in both sets.

Step 6 Preprocessing

Applied StandardScaler to numerical features and OneHotEncoder to categorical features via ColumnTransformer. The preprocessor was fitted only on training data to prevent data leakage.

Step 7 Model Training

Trained a Logistic Regression classifier (max_iter=1000, random_state=42) on the preprocessed training data.

Step 8 Evaluation

Evaluated the model using accuracy, precision, recall, F1 score, and a confusion matrix. Saved predictions to output/classification_outputs.csv.

5. Results

The model was trained on 304 samples and tested on 76 samples. The test set contained 72 No High Risk customers and 4 High Risk customers.

Performance Metrics

Metric	Score
Accuracy	93.42%
Precision	0.00%
Recall	0.00%
F1 Score	0.00%

Confusion Matrix

	Predicted: No Risk	Predicted: High Risk
Actual: No Risk	71 (True Negative)	1 (False Positive)
Actual: High Risk	4 (False Negative)	0 (True Positive)

The model correctly classified 71 out of 72 No High Risk customers but failed to correctly identify any of the 4 actual High Risk customers in the test set. Although accuracy appears high at 93.42%, precision, recall, and F1 score for the High Risk class are all 0%, showing that the model did not learn to detect the minority class.

6. Limitation and Recommendation

The main limitation is severe class imbalance. Only 5.3% of customers are labelled as High Risk. Logistic Regression learned to predict No for almost every customer because that strategy produces the highest overall accuracy on imbalanced data. As a result, all 4 actual High Risk customers in the test set were missed.

Two approaches can address this in future work:

- **Class Weighting:** Setting `class_weight` to `balanced` inside Logistic Regression tells the model to penalise errors on the minority class more heavily, forcing it to learn patterns for High Risk customers.
- **Oversampling with SMOTE:** Synthetic Minority Oversampling Technique creates synthetic High Risk examples to balance the training data before model fitting, giving the model more examples of the rare class to learn from.